Non-Independence in Crossed Designs

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Non-independence in a crossed example.

	Conditions		
Participant	**Experimenter Absent**	**Experimenter Present**	***Participant Average***
Y_{hi}	7	8	7.5
Y_{hi}	5	5	5
Y_{hi}	6	6	6
Y_{hi}	7	9	8
Y_{hi}	8	8	8
Y_{hi}	7	7	7
Y_{hi}	5	6	5.5
Y_{hi}	6	8	7
Condition Means	6.375	7.125	6.75
Grand Mean	6.75		

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Condition Means	6.375	7.125	6.75
Grand Mean	6.75		

Now, h = *condition*, i = *participant*.

Source	SS	df
Total Between Subjects	18.00	7
-- Condition	2.25	1
-- Error Within	2.75	7
Total Within Subjects	5.00	8
Total	23.00	15

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$$\sum \left(Y_{hi} - \bar{\bar{Y}}\right)^2$$

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$$\sum(\bar{Y}_i - \bar{\bar{Y}})^2 n_i$$

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NOTE THAT I ACCIDENTALLY LEFT "n" OUT OF THE SLIDES IN THE VIDEO.

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$$\sum\left((Y_{hi}-\bar{Y}_i)-\overline{(Y_{hi}-\bar{Y}_i)_h}\right)^2$$

$$\sum\left(diff_{hi}-\overline{diff_h}\right)^2$$

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$$\sum\left((\widehat{Y_{hi}-\bar{Y}_i}) - \overline{\overline{(Y_{hi}-\bar{Y}_i)}}\right)^2$$

$$\sum\left(\widehat{diff}_{hi} - \overline{\overline{diff}}\right)^2$$

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Our calculation of the MSR and MSE, and subsequently the F_observed, now relies on this SSE(A) with the between subjects variation removed.